

VehicleClassAttention: Learning PCU-Weighted Signal Control for Heterogeneous Indian Urban Traffic

Shrey Parekh^{1,*} and Mohini Reddy²

¹ Student, Computer Engineering, Mukesh Patel School of Technology Management and Engineering, SVKM's NMIMS, Mumbai, India

`shreyparekh3@gmail.com`

* Corresponding author

² Assistant Professor, Computer Engineering, Mukesh Patel School of Technology Management and Engineering, SVKM's NMIMS, Mumbai, India

`mohini.reddy5@gmail.com`

Abstract. Adaptive traffic signal control in Indian cities is complicated by heterogeneous vehicle compositions where two-wheelers, auto-rickshaws, cars, and buses share road space with widely different Passenger Car Unit (PCU) equivalents. Existing reinforcement learning approaches treat all vehicles uniformly, ignoring class-level differences that dominate Indian urban traffic. This paper introduces VehicleClassAttention (VCA), a learnable attention module that weights vehicle classes by their PCU-scaled congestion contribution and is initialised with IRC:106-1990 priors. We integrate VCA into a graph attention network for multi-intersection signal control and evaluate four model variants in a systematic ablation on a SUMO-simulated 9-intersection grid calibrated with Indian mixed-traffic parameters (60% two-wheelers, 16% auto-rickshaws, 18% cars, 6% buses). GAT-DQN with VCA achieves an average queue length of 10.57 ± 0.04 PCU, a 14.9% reduction over the best classical baseline (Fixed-Time, 12.42 PCU), with 5.7% higher throughput. The ablation shows graph structure accounts for the largest gain (12.5%), attention over convolution adds 1.8%, and VCA contributes a further 5.5%, confirming that explicit vehicle-class modelling improves signal control for heterogeneous Indian traffic.

Keywords: Deep reinforcement learning · Graph attention network · Indian mixed traffic · Passenger car unit · Traffic signal control · Vehicle class attention

1 Introduction

1.1 Motivation

Traffic signal control at urban intersections directly affects congestion, travel time, and vehicle emissions. In Indian cities, the problem is harder than in most other contexts because the traffic mix is heterogeneous:

two-wheelers ($\approx 60\%$), auto-rickshaws (16%), cars (18%), and buses (6%) share road space with very different physical footprints [13]. IRC:106-1990 quantifies these differences through Passenger Car Unit (PCU) equivalents (0.5, 0.75, 1.0, 3.0 respectively), so a queue of ten buses (30.0 PCU) creates six times the congestion pressure of ten two-wheelers (5.0 PCU); classical controllers nonetheless use only aggregate PCU sums. Fixed-time controllers [4,15] use predetermined phase durations. Webster’s method [4] computes cycle lengths from arrival rates but remains static once deployed. MaxPressure [5] adapts to real-time queues and is throughput-optimal. None of these methods account for vehicle class. Deep reinforcement learning (DRL) has produced strong results [1,6,10,14], with multi-intersection extensions using graph neural networks [9,11] and graph attention [7,8]. The observation spaces in [6,7,8] encode queues as raw vehicle counts or aggregate PCU values without distinguishing classes, a choice reflecting their car-dominated targets (Hangzhou, Jinan, New York). For Indian traffic with a 6:1 PCU ratio between bus and two-wheeler, this uniform treatment leads to suboptimal phase decisions.

1.2 Main Contributions and Organisation

We make three contributions: (i) **VehicleClassAttention (VCA)**, a learnable attention module that takes per-class vehicle counts and produces a PCU-weighted embedding for each intersection, with attention weights initialised from IRC:106-1990 and refined during training; (ii) **GAT-DQN integration**: VCA embedded in a graph attention architecture so that each intersection node attends to neighbours through topology-masked multi-head attention; and (iii) a **systematic ablation** of four variants (DQN, GNN-DQN, GAT-DQN-Base, GAT-DQN) against three classical baselines (Fixed-Time, Webster, MaxPressure) on a SUMO-simulated [12] 9-intersection grid with the IRC mixed-traffic mix. Across 3 seeds, GAT-DQN reaches 10.57 ± 0.04 PCU, a 14.9% queue reduction over Fixed-Time; the ablation isolates contributions of 12.5% (graph), 1.8% (attention over GCN) and 5.5% (VCA), with very low cross-seed variance for the graph-based models (std 0.03–0.15 PCU vs. 0.90 PCU for DQN).

The remainder of the paper is organised as follows. Section 2 reviews related work, Section 3 formalises the Dec-POMDP, Section 4 presents VCA and the GAT-DQN architecture, Section 5 describes the experiments, Section 6 reports results with a comparison to prior work, and Section 7 concludes.

2 Related Work

Classical signal control. Fixed-time plans remain the most common scheme at Indian intersections [15]; Webster [4] optimises cycle length from measured flow but cannot adapt online, and Varaiya [5] formalised the throughput-optimal MaxPressure controller. None learn class-specific importance.

RL for signal control. Independent DQN agents have been applied at single [10,1] and multi-intersection [11] scales, with phase-sensitive state design [6], MaxPressure reward shaping [7], and phase-symmetry encoding [16]; surveys in [14,17] consolidate the design space.

Graph neural networks. GCNs [2] were applied to signal control by Nishi et al. [9]; graph attention networks [3] were used by CoLight [8] on networks of up to 196 intersections. All operate on uniform vehicle representations.

Heterogeneous traffic. Existing RL controllers [6,7,8] target car-only networks (Hangzhou, Jinan, New York), where standard level-of-service metrics are known to break down for Indian mixed traffic. IRC:106-1990 [13] addresses heterogeneity through static PCU equivalence factors; Arasan and Krishnamurthy [18] reported that PCU values shift with congestion and lateral spacing, which a static lookup cannot capture. To our knowledge, no existing DRL signal controller incorporates a learnable mechanism for weighting vehicle classes. Our VCA module fills this gap by training PCU-initialised attention through the Q-network gradient.

3 System Model and Problem Formulation

3.1 Traffic Network

We model the road network as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ is the set of signalised intersections and $\mathcal{E}$ is the set of road segments connecting them. In our setup, $N = 9$ intersections are arranged in a 3×3 grid (Fig. 1). Each intersection v_i is connected to its horizontally and vertically adjacent neighbours. Corner intersections have 2 neighbours, edge intersections have 3, and the centre intersection has 4. The adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$ captures this connectivity: $A_{ij} = 1$ if intersections v_i and v_j are connected by a road segment, $A_{ij} = 0$ otherwise. Each road segment carries bidirectional traffic along either the North-South (NS) or East-West (EW) direction.

3.2 Mixed Traffic Model

Vehicles are injected by a Poisson process with base rate $\lambda = 0.19$ per route per step (6 NS + 6 EW routes, ≈ 2.28 vehicles/step). Each vehicle is sampled from the IRC:106-1990 mix [13] introduced in Section 1.1 (60% two-wheelers, 16% auto-rickshaws, 18% cars, 6% buses with PCU 0.5/0.75/1.0/3.0); the mix-weighted average PCU is 0.78, so the average vehicle occupies less road space than a car.

3.3 Signal Control as a Markov Decision Process

We formulate the problem as a decentralised partially observable Markov decision process (Dec-POMDP) with parameter sharing: one policy π_θ is shared across all N intersections. Each intersection v_i observes a 24-dimensional vector $\mathbf{o}_i^t \in \mathbb{R}^{24}$ comprising NS/EW vehicle counts and PCU

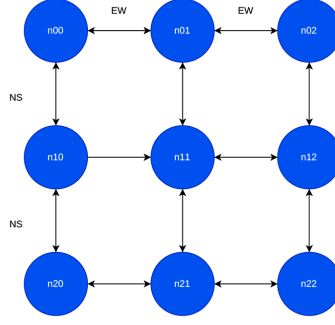


Fig. 1. The 3×3 grid network topology used in the experiments. Each circle represents a signalised intersection. Vertical edges carry NS traffic and horizontal edges carry EW traffic. The adjacency structure determines the graph attention neighbourhood for each intersection.

queues, current phase, steps since last switch, NS and EW per-class counts (8 features), a scenario flag, six neighbour aggregates (mean/max NS/EW PCU and neighbour phase fractions), an action mask, and NS/EW queue derivatives. Graph models process the full matrix $\mathbf{O}^t \in \mathbb{R}^{N \times 24}$; the DQN baseline sees only its local vector. The action $a_i^t \in \{0, 1\}$ is *keep* or *switch*; a switch requires the current green to have been active for $g_{\min} = 5$ steps and triggers a $\tau_c = 2$ -step all-red clearance before transitioning between NS_G and EW_G. Each agent’s reward combines queue minimisation with a pressure-based shaping term:

$$r_i^t = -\frac{q_i^{\text{NS}} + q_i^{\text{EW}}}{\eta} + \alpha \cdot \max\left(\frac{p_i^t}{\eta}, 0\right), \quad (1)$$

with $\eta = 30$, $\alpha = 0.2$, and pressure $p_i^t = q_i^{\text{NS}} - q_i^{\text{EW}}$ during NS_G (sign flipped during EW_G; zero during all-red). Graph models receive the network-average reward $\bar{r}^t = \frac{1}{N} \sum_i r_i^t$; DQN receives per-intersection rewards. Each agent maximises $J(\theta) = \mathbb{E}[\sum_{t=0}^T \gamma^t r_i^t]$ with $\gamma = 0.99$, $T = 300$, optimised via DQN [1] with experience replay and soft target updates.

4 Proposed Methodology

This section describes the architecture of our GAT-DQN model with VehicleClassAttention, the graph attention layers used for spatial coordination, the dataset generated through SUMO simulation, and the training procedure. Fig. 2 shows the end-to-end pipeline that links environment, observation extraction, the VCA-augmented GAT-DQN policy, action selection, and the off-policy training loop.

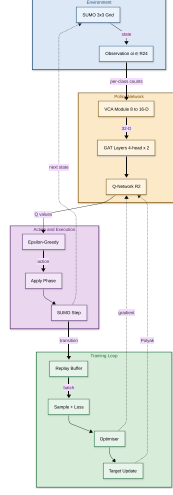


Fig. 2. End-to-end pipeline. Environment (blue) emits observations to the policy network (orange): VCA gates eight class counts into a 16-D embedding, augmented to a 32-D node vector, processed by two GAT layers and mapped to action values. The action stage (purple) applies an ε -greedy phase via SUMO; the transition is replayed by the training loop (green) to update θ and the Polyak target θ^- .

4.1 Dataset and Scenario Generation

Training data is generated by SUMO simulation since open Indian per-class count datasets at 1 Hz are not publicly available and the ablation requires controlled vehicle composition. SUMO is calibrated to the IRC mix [13] and the lateral filtering behaviour of [18]; each episode is a 300-step Poisson injection with a fixed route-generator seed so all variants see matched demand. Two-wheeler filtering is a stochastic 30% chance of a small speed boost (capped at 2.0 m/s) when the lane queue exceeds five vehicles. Across 250 episodes per seed each model sees $\approx 171k$ vehicle insertions, with the class distribution held within $\pm 1\%$ of the IRC mix by rejection sampling.

4.2 VehicleClassAttention (VCA)

VCA is a learnable attention module that produces a context embedding from per-class vehicle counts. At each intersection v_i , the 8 per-class counts at observation indices 6–13 are stacked as $\mathbf{x}_i^{\text{cls}} \in \mathbb{R}^8$, ordered two-wheeler, auto-rickshaw, car, and bus for NS followed by EW. A learnable PCU-prior vector $\mathbf{w}^{\text{pcu}} \in \mathbb{R}^8$ is initialised with the IRC:106-1990 values [0.5, 0.75, 1.0, 3.0, 0.5, 0.75, 1.0, 3.0] and applied through a sigmoid gate:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i^{\text{cls}} \odot \sigma(\mathbf{w}^{\text{pcu}}), \quad (2)$$

where σ keeps the scaling positive while allowing the prior to be refined during training. The gated input is passed through a two-layer attention

network with parameters $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{8 \times 8}$:

$$\alpha_i = \text{softmax}(\mathbf{W}_2 \tanh(\mathbf{W}_1 \tilde{\mathbf{x}}_i + \mathbf{b}_1) + \mathbf{b}_2), \quad (3)$$

producing attention weights that sum to 1. The weighted class features are projected to a 16-dimensional embedding with layer normalisation:

$$\mathbf{z}_i^{\text{vca}} = \text{LayerNorm}(\mathbf{W}_e(\alpha_i \odot \tilde{\mathbf{x}}_i) + \mathbf{b}_e), \quad \mathbf{W}_e \in \mathbb{R}^{16 \times 8}. \quad (4)$$

The 16-dimensional embedding replaces the 8 raw class counts in $\mathbf{o}_i$, producing an augmented 32-dimensional node feature vector $\hat{\mathbf{o}}_i \in \mathbb{R}^{32}$ that is input to the graph attention layers.

4.3 Graph Attention Network

The augmented features $\hat{\mathbf{O}} \in \mathbb{R}^{N \times 32}$ are processed by two graph attention layers [3]. In each layer the attention coefficient between intersections v_i, v_j is $e_{ij} = \text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\hat{\mathbf{o}}_i \parallel \mathbf{W}\hat{\mathbf{o}}_j])$, normalised over the neighbourhood $\mathcal{N}_i$ as $\alpha_{ij} = \exp(e_{ij}) / \sum_{k \in \mathcal{N}_i} \exp(e_{ik})$. Multi-head attention with $H = 4$ heads concatenates the outputs:

$$\hat{\mathbf{o}}'_i = \left\|_{h=1}^H \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^h \mathbf{W}^h \hat{\mathbf{o}}_j \right) \right\|. \quad (5)$$

The second GAT layer averages across heads instead of concatenating, producing the final node representation that is passed to the Q-value output layer.

4.4 Ablation Model Variants

To isolate the contribution of each architectural component we evaluate four variants. **DQN** is an independent per-intersection controller (3-layer MLP, no graph). **GNN-DQN** adds GCN-based neighbour aggregation [2]. **GAT-DQN-Base** replaces GCN with graph attention [3]. **GAT-DQN** (ours) adds VCA on top of graph attention. The pairwise comparisons isolate graph structure (DQN→GNN-DQN), attention over convolution (GNN-DQN→GAT-DQN-Base), and VCA (GAT-DQN-Base→GAT-DQN).

4.5 Training Procedure

All four models are trained with the DQN algorithm [1] using experience replay, soft target updates, and shared hyperparameters (Section 5). The full training loop is given in Algorithm 1.

Target updates use Polyak averaging $\theta^- \leftarrow \tau\theta + (1-\tau)\theta^-$ with $\tau = 0.005$ at every training step, which gave more stable training than hard target copies in preliminary runs. Exploration ε decays linearly from 1.0 to 0.01 across the 75,000 training steps and reaches its minimum around episode 120, leaving roughly 130 episodes of near-pure exploitation for evaluation. The independent DQN stores one transition per agent per step; graph models store one transition per environment step containing all 9 agents' joint observations, actions, and rewards.

Algorithm 1: GAT-DQN-VCA training.

Input: adjacency $\mathbf{A}$, $E = 250$, $T = 300$, $B = 256$, $\tau = 0.005$, PCU prior $\mathbf{w}_{\text{init}}^{\text{pcu}}$
Output: policy π_θ

```

1 Initialise  $Q_\theta$  with VCA  $f_{\text{VCA}}(\cdot; \mathbf{w}_{\text{init}}^{\text{pcu}})$ ;  $Q_{\theta^-} \leftarrow Q_\theta$ ;  $\mathcal{D} \leftarrow \emptyset$ 
2 for  $e \leftarrow 1$  to  $E$  do
3   Reset SUMO;  $\varepsilon \leftarrow \text{linear}(1.0, 0.01)$ 
4   for  $t \leftarrow 0$  to  $T - 1$  do
5      $\hat{\mathbf{o}}_i \leftarrow [\mathbf{o}_i^{0:6}, f_{\text{VCA}}(\mathbf{o}_i^t[6:13]), \mathbf{o}_i^{14:24}] \forall i$  // shared
6      $\mathbf{q} \leftarrow Q_\theta(\hat{\mathbf{O}}, \mathbf{A})$ ;  $a_i^t \leftarrow \varepsilon\text{-greedy} + \text{action mask}$ 
7     Step SUMO; push  $(\mathbf{O}^t, \{a_i^t\}, \bar{r}^t, \mathbf{O}^{t+1})$  to  $\mathcal{D}$ 
8     if  $|\mathcal{D}| \geq 1000$  then
9        $\mathcal{B} \sim \mathcal{D}$ ,  $|\mathcal{B}| = B$ ;  $y \leftarrow r + \gamma \max_{a'} Q_{\theta^-}(s', a')$ 
10       $\theta \leftarrow \theta - \nabla \text{SmoothL1}(Q_\theta(s, a), y)$ ;  $\theta^- \leftarrow \tau\theta + (1 - \tau)\theta^-$ 
11    end
12  end
13 end
```

5 Experimental Setup

5.1 Simulation and Scenario

Experiments use SUMO [12] with a 1 s step length, 300-step episodes, vehicle teleportation disabled, and lane-filtering calibrated to Indian intersections [18]. We evaluate on the uniform scenario (NS and EW receive equal demand) so that any performance difference is attributable to model architecture rather than directional bias. Each model is trained for 250 episodes across 3 random seeds (1, 2, 3).

5.2 Baseline Controllers

We compare against three classical controllers spanning the methods deployed at Indian intersections. **Fixed-Time** [15] cycles NS_G-RED-EW_G-RED with equal 30-second green phases. **Webster** [4] computes the optimal cycle length $C^* = (1.5L + 5)/(1 - Y)$ from estimated flow ratios with lost time $L = 8$ s; under uniform demand NS and EW receive equal splits. **MaxPressure** [5] switches when the opposing queue exceeds the served direction by more than 1.0 PCU; the same $g_{\min} = 5$ -step minimum-green constraint as the RL agents applies. All three baselines are evaluated over 15 episodes (5 episodes $\times$ 3 seeds) using the same SUMO environment and injection process as the RL models.

5.3 RL Model Configuration

Table 1 summarises the shared and model-specific hyperparameters used across all four RL model variants. All variants use vanilla DQN targets (no Double DQN) to keep training dynamics consistent across the ablation, and rewards are clipped to $[-1, 1]$ before computing the loss.

Table 1. Training hyperparameters, shared across all model variants except where noted.

Parameter	Value	Parameter	Value
<i>Shared across all models</i>			
Learning rate	0.001 (Adam)	Loss function	Smooth-L1 (Huber)
Discount γ	0.99	Epsilon decay	Linear over steps
Batch size	256	Epsilon start / end	1.0 / 0.01
Replay buffer	50,000	Episodes per seed	250
Min buffer (warm-up)	1,000	Steps per episode	300
Gradient clip norm	1.0	Target update	$\tau = 0.005$
<i>Signal control constraints (all models)</i>			
Minimum green time	5 steps	Action space	{keep, switch}
Clearance duration	2 steps		
<i>Graph models</i>		<i>DQN baseline</i>	
GAT attention heads	4	Hidden layers	3 (MLP)
GAT dropout	0.1	Reward type	Per-intersection
VCA embedding dim	16		
Reward type	Network-average		

5.4 Evaluation Protocol

Training runs in parallel with one SUMO instance per model, each on a unique network port. We evaluate using the last 30 episodes of each 250-episode run ($\varepsilon = 0.01$, near-pure exploitation). Two metrics are reported: **average queue length (PCU)**, the mean PCU-weighted queue across all 9 intersections averaged over all steps in an episode (lower is better), and **throughput**, the total number of vehicles completing their route per episode (higher is better). We report mean $\pm$ std across seeds. Improvements are computed relative to the best classical baseline (Fixed-Time at 12.42 PCU).

6 Results and Discussion

6.1 Overall Performance

Table 2 and Fig. 3 present the performance of all controllers on the uniform traffic scenario, averaged across 3 random seeds. The three classical baselines (Fixed-Time, Webster, MaxPressure) provide reference points, while the four RL model variants form the ablation chain.

Among the classical baselines, Fixed-Time achieves the lowest queue (12.42 PCU), beating Webster (13.45 PCU) and MaxPressure (14.18 PCU). MaxPressure’s reactive switching triggers frequent clearance phases under balanced demand where neither direction dominates, consistent with [5]: MaxPressure is throughput-optimal but not queue-optimal, and its edge over Fixed-Time appears under asymmetric demand. Among the RL models, GAT-DQN with VCA achieves 10.57 ± 0.04 PCU (a 14.9%

Table 2. Performance comparison on the uniform scenario (mean $\pm$ std across 3 seeds). Improvement is the queue reduction relative to Fixed-Time. Bold indicates best.

Controller	Queue (PCU)	Throughput	Impr.
<i>Classical baselines</i>			
Fixed-Time	12.42 ± 0.77	436 ± 32	–
Webster	13.45 ± 1.04	429 ± 27	–
MaxPressure	14.18 ± 0.92	389 ± 15	–
<i>RL models (last 30 ep., $\varepsilon = 0.01$)</i>			
DQN	13.02 ± 0.90	414 ± 15	–4.8%
GNN-DQN	11.39 ± 0.03	450 ± 3	+8.3%
GAT-DQN-Base	11.18 ± 0.15	449 ± 2	+10.0%
GAT-DQN	10.57 ± 0.04	461 ± 4	+14.9%

Baselines: 15 episodes (5×3 seeds). RL: last 30 episodes per seed, aggregated over 3 seeds (90 total).

Table 3. Ablation: per-component queue reduction. ΔQ is the change in mean queue length (PCU); reduction is the percentage relative to the source model.

From $\rightarrow$ To	Added	ΔQ	Reduction
DQN $\rightarrow$ GNN-DQN	GCN graph	–1.63	12.5%
GNN-DQN $\rightarrow$ GAT-DQN-Base	Attention	–0.21	1.8%
GAT-DQN-Base $\rightarrow$ GAT-DQN	VCA	–0.61	5.5%
DQN $\rightarrow$ GAT-DQN	All	–2.45	18.8%

reduction over Fixed-Time) and the highest throughput (461 vehicles, +5.7%); each architectural addition reduces queue length over the previous variant. The graph-based models show low cross-seed variance (std 0.03–0.15 PCU) compared to DQN (std 0.90 PCU), indicating that graph structure produces policies less sensitive to random initialisation in addition to lowering mean queue length.

6.2 Ablation Analysis

Table 3 isolates the contribution of each component by measuring the queue reduction at each step of the ablation chain.

Adding graph structure (DQN $\rightarrow$ GNN-DQN) gives the largest single improvement, 1.63 PCU (12.5%): an intersection that can observe upstream queues adjusts phase timing to accommodate incoming platoons. The independent DQN lacks this capability and its queue length actually rises from exploration (~ 10.4 PCU early) to exploitation (13.02 PCU last 30

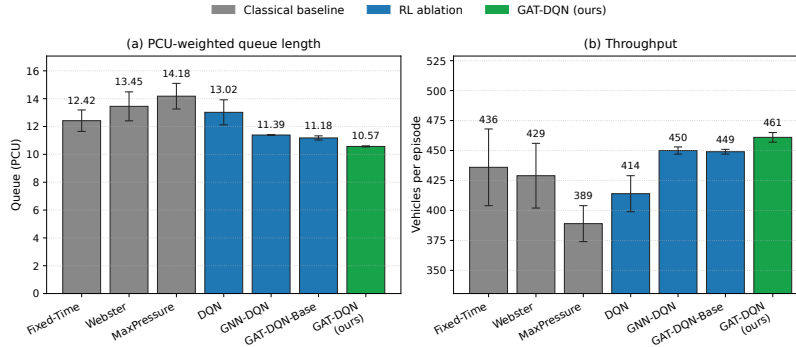


Fig. 3. Final-30-episode performance for all controllers on the uniform scenario, with ± 1 standard deviation error bars across 3 seeds. (a) PCU-weighted queue length: GAT-DQN with VCA produces the lowest mean queue (10.57 PCU) with negligible cross-seed variance. (b) Throughput: the same model produces the highest completion rate (461 vehicles per episode).

episodes), with a $30\times$ higher cross-seed std (0.90 vs. 0.03 PCU). Replacing GCN with attention (GNN-DQN $\rightarrow$ GAT-DQN-Base) adds 0.21 PCU (1.8%); the gain is modest on a regular grid where all neighbours have similar structural roles and would likely be larger on irregular real-world networks. Adding VCA (GAT-DQN-Base $\rightarrow$ GAT-DQN) reduces queue length by 0.61 PCU (5.5%) at a cost of only 400 additional parameters. The VCA gain reflects the non-linear mapping between count and congestion: a queue of 10 vehicles in our mix can range from 5.0 PCU (all two-wheelers) to 30.0 PCU (all buses). With class standard deviation $\sigma_c = 0.59$ PCU, the PCU uncertainty from ignoring class scales as $\sqrt{n}\sigma_c$ and reaches 2.6–3.2 PCU at queues of 20–30 vehicles, so the 5.5% improvement we measure at moderate load is a lower bound that grows with congestion intensity.

6.3 Training Dynamics

Fig. 4 shows the queue length progression during training for all four models, smoothed with a 15-episode exponential moving average and shaded with the rolling 15-episode standard deviation. All graph-based models (GNN-DQN, GAT-DQN-Base, GAT-DQN) converge to stable queue levels by approximately episode 80–100, while DQN exhibits higher variance and a rising queue trend during the exploitation phase.

6.4 Comparison with Existing Literature

Prior RL controllers (IntelliLight [6], PressLight [7], CoLight [8], FRAP [16], GCN-Signal [9]) report 8–19% queue reductions over their classical baselines [14,17] on car-only networks; our 14.9% on a 3×3 IRC-mix grid falls in the upper half of that range and is the only entry whose evaluation

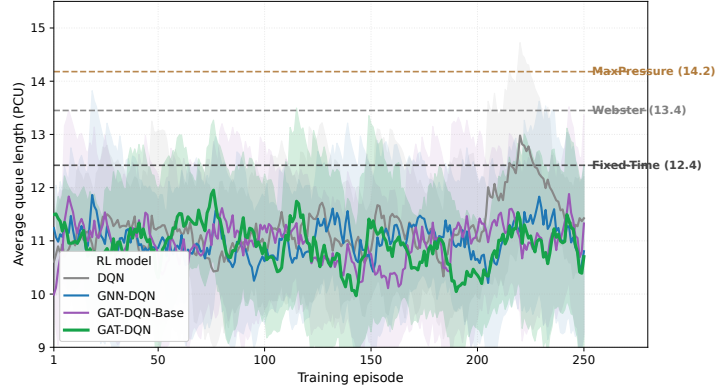


Fig. 4. Average queue length (PCU) over 250 training episodes (seed 3). Curves: 15-episode EMA; bands: rolling 15-episode std. Dashed lines are classical baselines. Graph models converge below Fixed-Time by episode 50; DQN degrades during exploitation. Aggregate cross-seed statistics are in Table 2.

includes per-class heterogeneity, with the PCU reduction translating to a larger raw-vehicle reduction since the mix-average vehicle is sub-PCU (0.78).

7 Conclusion

We introduced VehicleClassAttention (VCA), a learnable attention module that weights vehicle classes by their PCU-scaled contribution to congestion, initialised from IRC:106-1990 priors and refined through reinforcement learning, and integrated into a graph attention network. On a SUMO-simulated 9-intersection grid with the IRC mix, GAT-DQN with VCA reached 10.57 ± 0.04 PCU (14.9% reduction over Fixed-Time) with 5.7% higher throughput across 3 seeds; the ablation isolates 12.5% (graph), 1.8% (attention), and 5.5% (VCA), and graph-based models show very low cross-seed variance (std 0.03–0.15 PCU).

Limitations and future work. The evaluation uses a synthetic 3×3 grid under uniform demand and lacks a linear-projection control for the augmented feature dimension, so part of the 5.5% VCA gain may come from capacity rather than attention. Future work: real Indian-city topologies via OpenStreetMap-to-SUMO, asymmetric peak-hour demand, and visualisation of learned attention weights to verify recovery of the expected class ordering.

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